

ETHIO-RAFFLE PLATFORM

NLA & MOR COMPLIANT

System Provider Model

Simplified System Architecture & Operational Guide (SRS v1.0.0)

1. PLATFORM IDENTITY (SYSTEM PROVIDER MODEL)

Who We Are: Ethio-Raffle is strictly a **Technology Infrastructure & Escrow Provider**. The platform owns zero physical inventory. We provide a trusted, licensed bridge connecting independent merchants (Sellers) with ticket buyers, backed by automated escrow protection and statutory government tax deduction.

2. HOW IT WORKS FOR ALL 3 PARTIES

 **Ticket Buyers**

- **No registration required** — purchase with mobile phone number.
- Instant ticket confirmation via SMS.
- Pay via Telebirr, CBE Birr, Chapa, Awash, or Agent Kiosks.
- **Winner Choice:** Physical item delivery OR Instant cash to wallet.

 **Verified Merchants**

- Must register with **Fayda National ID & Business License**.
- Post items with **3 distinct photos &** fair market value.
- Protected by **Escrow**: funds unlocked instantly upon Winner QR scan.
- 7-Day Auto-Release safety rule.

 **Platform Admins**

- **Stealth Access:** Triggered exclusively via Ctrl + Shift + A.
- Gate 1: Verify & approve Seller Fayda IDs.
- Gate 2: Anti-Gouging appraisal & counter-offers.
- Audit statutory 15% VAT & authorize seller cashouts.

3. 100% ANTI-CHEAT GUARANTEE (SHA-256 EXPLAINED SIMPLY)

To make it mathematically impossible for admins or sellers to rig the draw, the system uses a "Digital Wax Seal":

Step 1

Sealed in Advance
Before ticket #1 is sold, a secret key is locked into a public SHA-256 fingerprint (e.g. c9a9d30d...).

Step 2

Tamper-Proof Sales
Tickets #1 to #10,000 are sold. Changing even 1 letter in the secret completely changes the hash, exposing any cheat.

Step 3

Midnight Draw (12:00 AM)
When 100% sold out, the computer calculates the winner using pure math:
 $(Seed + Tickets) \% Total = Winner$.

Step 4

Public 1-Click Verification
Anyone can test the draw on the **/verifier** tool to confirm the math was 100% honest.

4. WINNER SETTLEMENT & HANDOVER PROTECTION

Stage	What Happens	Security & Financial Safeguard
1. Winner Notification	Winner receives SMS notification and accesses settlement console on /my-tickets.	Verified phone number authentication gate.
2. Choice of Prize	Option 1: Physical Item (Car/Phone/Condo) Option 2: Cash Equivalent credited to wallet.	If Cash is chosen, funds credit instantly to wallet; item stays in merchant stock.
3. Physical QR Handover	Winner presents Dynamic Claim QR Code to the merchant during physical pickup.	Merchant scans QR $\rightarrow$ Escrow balance instantly unfreezes into merchant payout balance.
4. 7-Day Safety Fallback	If QR scan is not completed, system auto-releases funds to merchant after 7 days.	If buyer reports non-receipt, escrow is permanently frozen for Admin Arbitration.

5. FINANCIAL SAFETY, ESCROW & STATUTORY 15% VAT

Revenue Stream	Percentage / Rate	Routing & Escrow Flow
Statutory Ethiopian VAT	15% of Ticket Price	Deducted at source on every ticket $\rightarrow$ Segregated into Government Tax Escrow (MoR compliance).
Seller Net Proceeds	85% Net Escrow	Locked in escrowBalance until physical QR scan or 7-day auto-release $\rightarrow$ payoutBalance.
Seller Cashout Gate	100% Admin Authorized	Seller requests withdrawal $\rightarrow$ Admin verifies and authorizes bank/Telebirr disbursement.